

Sheet 4 Lab Report

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1 Problem8

Schema Creation

Schema Requirements:

- Customer (cust_id, cust_name, annual_revenue, cust_type)
Cust_id must be between 100 and 10,000. Cust_type must be 'MANUFACTURER', 'WHOLE-SALER', or 'RETAILER'.
- Shipment (shipment_no, cust_id, weight, truck_no, destination, ship_date)
Foreign keys: cust_id references customer on delete cascade, truck_no references truck on delete set null, destination references city. Weight must be under 1000.
- Truck (truck_no, driver_name)
- City (city_name, population)

```
-- DDL Schema Creation
CREATE TABLE IF NOT EXISTS Customer (
  cust_id INTEGER PRIMARY KEY CHECK (
    cust_id >= 100
    AND cust_id <= 10000
  ),
  cust_name VARCHAR(30),
  annual_revenue NUMERIC(10, 2),
  cust_type VARCHAR(30) CHECK (
    cust_type IN ('MANUFACTURER', 'WHOLESALE', 'RETAILER')
  )
);

CREATE TABLE IF NOT EXISTS Truck (
  truck_no INTEGER PRIMARY KEY,
  driver_name VARCHAR(30)
);

CREATE TABLE IF NOT EXISTS City (
  city_name VARCHAR(30) PRIMARY KEY,
  population INTEGER
);

CREATE TABLE IF NOT EXISTS Shipment (
  shipment_no INTEGER PRIMARY KEY,
  cust_id INTEGER REFERENCES Customer (cust_id) ON DELETE CASCADE,
  weight INTEGER CHECK (weight < 1000),
  truck_no INTEGER REFERENCES Truck (truck_no) ON DELETE SET NULL,
```

```
destination VARCHAR(30) REFERENCES City (city_name)
);
```

Q1: Give names of customer who have sent packages (shipments) to Kolkata, Chennai and Mumbai.

```
SELECT
    cust_name
FROM
    customer
    JOIN shipment ON customer.cust_id = shipment.cust_id
WHERE
    destination in ('Kolkata', 'Chennai', 'Mumbai');
```

Output:

cust_name
Acme Corp
Acme Corp
Acme Corp
Echo Supplies
Echo Supplies
Echo Supplies
Halo Tech

Q2: List the names of the driver who have delivered shipments weighing over 200 pounds.

```
SELECT
    driver_name ,
    weight
FROM
    truck
    JOIN shipment on truck.truck_no = shipment.truck_no
WHERE
    shipment.weight > 200;
```

Output:

driver_name	weight
IQBAL	300
MANISH	210
DEV	500
RAJ	250
AMIT	300
IQBAL	999
IQBAL	750

Q3: Retrieve the maximum and minimum weights of the shipments. Rename the output as Max_Weight and Min_Weight respectively.

```
SELECT
    MAX(weight) as Max_Weight ,
    MIN(weight) as Min_Weight
FROM
    shipment;
```

Output:

max_weight	min_weight
999	10

Q4: For each customer, what is the average weight of package sent by the customer?

```
SELECT
    cust_name ,
    ROUND(AVG(weight), 2)
FROM
    customer
    JOIN shipment on customer.cust_id = shipment.cust_id
GROUP BY
    customer.cust_id
ORDER BY
    AVG(weight) DESC;
```

Output:

cust_name	round
Jupiter Mills	625.00
Gamma Traders	544.50
Indus Retail	210.00
Beta Goods	200.00
Echo Supplies	180.00
Acme Corp	150.00
Delta Dynamics	120.00
Charlie Retail	116.67
Foxtrot Mart	80.00
Halo Tech	52.50

Q5: List the names and populations of cities that have received a shipment weighing over 100 pounds.

```
SELECT
    city_name ,
    population ,
    weight
FROM
    city
JOIN shipment ON shipment.destination = city.city_name
WHERE
    weight > 100
ORDER BY
    weight ASC;
```

Output:

city_name	population	weight
Mumbai	12000000	150
Kolkata	4500000	150
Chennai	7000000	150
Pune	800000	150
Delhi	11000000	150
Bangalore	8000000	150
Hyderabad	6900000	150
Jaipur	3000000	150
Surat	4400000	150
Kochi	600000	150
Pune	800000	210
Chennai	7000000	250
Pune	800000	300
Kolkata	4500000	300
Pune	800000	500
Kochi	600000	750
Delhi	11000000	999

Q6: List cities that have received shipments from every customer.

```
SELECT
    city.city_name
FROM
    city
    JOIN shipment on shipment.destination = city.city_name
GROUP BY
    city.city_name
HAVING
    COUNT(DISTINCT shipment.cust_id) = (
        SELECT
            COUNT(cust_id)
        FROM
            customer
    );
```

Output:

city_name
Pune

Q7: For each city, what is the maximum weight of a package sent to that city?

```
SELECT
    city.city_name ,
    MAX(shipment.weight)
FROM
    city
    JOIN shipment on city.city_name = shipment.destination
GROUP BY
    city.city_name;
```

Output:

city_name	max
Kochi	750
Hyderabad	150
Kolkata	300
Pune	500
Jaipur	150
Mumbai	150
Delhi	999
Surat	150
Chennai	250
Bangalore	150

Q8: List the name and annual revenue of customers whose shipments have been delivered by truck driver 'IQBAL'.

```
SELECT DISTINCT
    customer.cust_name ,
    customer.annual_revenue
FROM
    customer
    JOIN shipment on customer.cust_id = shipment.cust_id
    JOIN truck on shipment.truck_no = truck.truck_no
WHERE
    driver_name = 'IQBAL';
```

Output:

cust_name	annual_revenue
Beta Goods	300000.00
Echo Supplies	200000.00
Gamma Traders	450000.00
Jupiter Mills	600000.00

Q9: List drivers who have delivered shipments to every city.

```
SELECT
    truck.driver_name
FROM
    truck
    JOIN shipment on shipment.truck_no = truck.truck_no
GROUP BY
    truck.truck_no
HAVING
    COUNT(DISTINCT shipment.destination) = (
        SELECT
            COUNT(city_name)
        FROM
            city
    );
```

Output:

driver_name
SAM

Q10: For each city, with population over 1 million, what is the minimum weight of a package sent to that city.

```
SELECT
    city.city_name,
    MIN(shipment.weight)
FROM
    city
    JOIN shipment on shipment.destination = city.city_name
WHERE
    city.population > 1000000
GROUP BY
    city.city_name
```

Output:

city_name	min
Mumbai	10
Jaipur	150
Delhi	150
Surat	150
Chennai	150
Hyderabad	150
Bangalore	150
Kolkata	150

2 Problem9

Schema Creation

Schema Requirements:

- EMP (EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO)
EMPNO must be between 7000 and 8000. ENAME must not exceed 10 characters. JOB must be in ('Clerk', 'Salesman', 'Manager', 'Analyst', 'President'). MGR is the manager's EMPNO. COMM must be under 1500 and defaults to 0.
- DEPT (DEPTNO, DNAME, LOC)
DEPTNO must start with 'D'. DNAME must be 'Accounting' or 'Sales' or 'Research' or 'Operations'.

```
-- DDL Schema Creation
DROP TABLE IF EXISTS EMP CASCADE;

DROP TABLE IF EXISTS DEPT CASCADE;

CREATE TABLE IF NOT EXISTS dept (
  deptno VARCHAR(10) PRIMARY KEY CHECK (DEPTNO LIKE 'D%'),
  dname VARCHAR(20) CHECK (
    dname IN ('Accounting', 'Sales', 'Research', 'Operations')
  ),
  loc VARCHAR(30)
);

CREATE TABLE IF NOT EXISTS emp (
  empno INTEGER PRIMARY KEY CHECK (
    empno >= 7000
    AND empno <= 8000
  ),
  ename VARCHAR(10),
  job VARCHAR(15) CHECK (
    job IN (
      'Clerk',
      'Salesman',
      'Manager',
      'Analyst',
      'President'
    )
  ),
  mgr INTEGER REFERENCES EMP (empno) ON DELETE SET NULL,
  hiredate DATE,
  sal NUMERIC(10, 2),
  comm NUMERIC(10, 2) DEFAULT 0 CHECK (comm < 1500),
  deptno VARCHAR(10) REFERENCES DEPT (DEPTNO)
);
```

Q1: Display the difference between highest and lowest salary of each department in descending order. Label the column as “Difference”.

```
SELECT
    dname,
    MAX(sal) - MIN(sal) AS Difference
FROM
    dept
    JOIN emp on emp.deptno = dept.deptno
GROUP BY
    dname
ORDER BY
    MAX(sal) - MIN(sal) DESC;
```

Output:

dname	difference
Accounting	3700.00
Research	2200.00
Sales	1900.00

Q2: List all the employees' employee number and name along with their immediate managers' employee number and name.

```
SELECT
    e1.empno ,
    e1.ename ,
    e2.empno as manager_id ,
    e2.ename as manager_name
FROM
    emp e1
    LEFT JOIN emp e2 on e1.mgr = e2.empno ;
```

Output:

empno	ename	manager_id	manager_name
7839	Andrew	NULL	NULL
7566	JONES	7839	Andrew
7698	BLAKE	7839	Andrew
7782	CLARK	7839	Andrew
7788	SCOTT	7566	JONES
7902	FORD	7566	JONES
7499	ALLEN	7698	BLAKE
7521	WARD	7698	BLAKE
7654	MARTIN	7698	BLAKE
7369	SMITH	7902	FORD
7876	ADAMS	7788	SCOTT
7900	JAMES	7698	BLAKE
7934	MILLER	7782	CLARK

Q3: Create a query that will display the total number of employees and the total number of employees who were hired only in 2020. Give the column headings as “TOTAL” and “TOTAL_2020” respectively.

```
SELECT
(
  SELECT
    COUNT(*)
  FROM
    emp
) AS TOTAL,
(
  SELECT
    COUNT(*)
  FROM
    emp
  WHERE
    EXTRACT(
      YEAR
    FROM
      hiredate
    ) = 2020
) as TOTAL_2020;
```

Output:

total	total_2020
13	8

Q4: Display the manager number and the salary of the lowest paid employee under that manager. Exclude anyone whose manager is not known. Exclude any group where the minimum salary is less than 1000. Sort the output in descending order of salary.

```
SELECT
  mgr as manager_id,
  MIN(sal) as min_sal
FROM
  emp
WHERE
  mgr IS NOT NULL
GROUP BY
  mgr
HAVING
  MIN(sal) >= 1000
ORDER BY
  min_sal DESC;
```

Output:

manager_id	min_sal
7566	3000.00
7839	2450.00
7782	1300.00
7788	1100.00

Q5: Assume that there are some departments where no employee is assigned. Now, write a query to display the department name, location name, number of employees, and the average salary for all the employees in that department. Label the columns as “DNAME”, “LOCATION”, “NUMBER OF PEOPLE”, and “AVERAGE SALARY” respectively. Round the average salary to two decimal places. The outcome of the query must include the details of the departments where no employee is assigned and in that case the “AVERAGE SALARY” for that department is to be displayed as 0 (zero).

```
SELECT
    dname,
    loc AS LOCATION,
    COUNT(empno) as "NUMBER OF PEOPLE",
    ROUND(COALESCE(AVG(sal), 0), 2) as "AVERAGE SALARY"
FROM
    dept d
    LEFT JOIN emp e ON d.deptno = e.deptno
GROUP BY
    dname,
    loc
ORDER BY
    "AVERAGE SALARY" DESC;
```

Output:

dname	location	NUMBER OF PEOPLE	AVERAGE SALARY
Accounting	NEW YORK	3	2916.67
Research	DALLAS	5	2175.00
Sales	CHICAGO	5	1580.00
Operations	BOSTON	0	0.00